(i) Let $u = 2 - x,\ du = -dx$

$$\begin{aligned}\int \ln(2-x)\,dx &= -\int \ln u\,du \\ &= -(u\ln u - u) + C \\ &= -(2-x)\ln(2-x) + (2-x) + C\end{aligned}$$

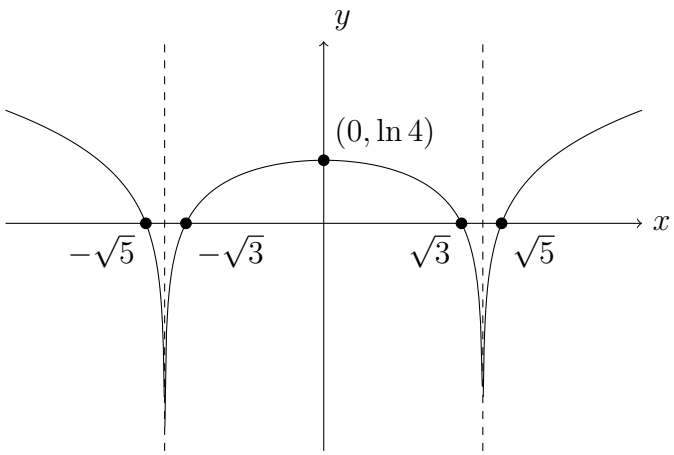
(ii)

$$y = \ln|x^2 - 4|$$

$$|x^2 - 4| = 1 \implies x = \pm\sqrt{3},\ \pm\sqrt{5}$$

Vertical asymptotes: $x = \pm 2$

x	$-\sqrt{5}$	-2	$-\sqrt{3}$	0	$\sqrt{3}$	2	$\sqrt{5}$
y	0	$-\infty$	0	$\ln 4$	0	$-\infty$	0



(iii) In the first quadrant,

$$y = \ln(4 - x^2)$$

The curve meets the positive x-axis when

$$\ln(4 - x^2) = 0 \implies x = \sqrt{3}$$

$$
\begin{aligned}
\text{Area} &= \int_0^{\sqrt{3}} \ln(4-x^2)\,dx \\
&= \Big[x\ln(4-x^2)\Big]_0^{\sqrt{3}} + 2\int_0^{\sqrt{3}} \frac{x^2}{4-x^2}\,dx \\
&= 2\int_0^{\sqrt{3}} \left(-1 + \frac{4}{4-x^2}\right)\,dx \\
&= \left[-2x + 2\ln\left(\frac{2+x}{2-x}\right)\right]_0^{\sqrt{3}} \\
&= -2\sqrt{3} + 2\ln\left(\frac{2+\sqrt{3}}{2-\sqrt{3}}\right) \\
&= -2\sqrt{3} + 4\ln\left(2+\sqrt{3}\right)
\end{aligned}
$$

(iv)

$$
\begin{aligned}
\int_0^2 \ln(4-x^2)\,dx &= \Big[-(2-x)\ln(2-x) + (2+x)\ln(2+x) - 2x\Big]_0^2 \\
&= 4\ln 4 - 4
\end{aligned}
$$

$\because \ln(4-x^2) \geq 0$ on $0 \leq x \leq \sqrt{3}$ and $\ln(4-x^2) < 0$ on $\sqrt{3} < x < 2$,

$$
\begin{aligned}
\therefore \text{Area} &= 2\left(\int_0^{\sqrt{3}} \ln(4-x^2)\,dx - \int_{\sqrt{3}}^2 \ln(4-x^2)\,dx\right) \\
&= 2\left(2\int_0^{\sqrt{3}} \ln(4-x^2)\,dx - \int_0^2 \ln(4-x^2)\,dx\right) \\
&= 4\left(4\ln(2+\sqrt{3}) - 2\sqrt{3}\right) - 2(4\ln 4 - 4) \\
&= 8 + 16\ln(2+\sqrt{3}) - 16\ln 2 - 8\sqrt{3}
\end{aligned}
$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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